

GUYMON, OK, USA

WMO: 722095

Lat: 36.682N

Lon: 101.505W

Elev: 3123

StdP: 13.11

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	8.8	13.5	-2.1	5.5	19.2	2.1	6.9	26.8	34.1	43.1	30.8	39.4	8.9	10	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	101.4	67.8	99.0	67.5	96.3	67.3	72.4	89.4	71.3	88.9	70.4	88.1	14.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	115.4	76.5	66.6	110.1	75.7	65.7	106.4	75.3	38.1	89.3	37.1	88.9	36.2	88.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.2	26.5	23.9	DB	0.2	106.2	5.2	2.8	-3.5	108.2	-6.5	109.9	-9.4	111.4	-13.2	113.5
			WB	-0.9	74.6	4.9	1.1	-4.4	75.4	-7.3	76.0	-10.1	76.6	-13.7	77.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	36.8	39.1	48.7	56.4	66.1	77.0	80.6	78.3	71.8	58.4	46.7	36.5	(6)
	DBStd	18.06	10.06	11.02	9.81	9.14	8.64	6.02	5.17	5.40	7.04	9.18	9.58	9.89	(7)
	HDD50	1556	419	329	139	41	3	0	0	0	0	30	169	426	(8)
	HDD65	4182	875	727	509	281	93	1	0	0	23	240	549	884	(9)
	CDD50	4525	9	23	99	233	501	810	948	879	653	292	70	8	(10)
	CDD65	1676	0	1	4	23	126	361	484	414	226	36	1	0	(11)
	CDH74	21923	4	43	220	637	1918	4610	6226	4802	2679	689	93	2	(12)
	CDH80	11698	0	10	60	241	899	2554	3682	2659	1319	259	15	0	(13)
(14) Wind	WSAvg	10.9	10.0	10.8	11.8	13.1	11.7	12.0	10.4	9.5	10.8	10.6	10.6	10.0	(14)
(15) Precipitation	PrecAvg	18.3	0.5	0.5	1.1	1.4	2.2	2.9	3.0	3.0	1.2	1.7	0.5	0.6	(15)
	PrecMax	21.8	1.7	1.5	4.5	4.1	4.8	6.9	6.5	5.6	3.1	6.0	3.5	3.9	(16)
	PrecMin	13.7	0.0	0.2	0.0	0.1	0.4	0.3	0.9	0.4	0.1	0.0	0.0	0.0	(17)
	PrecStd	2.0	0.5	0.4	1.0	1.0	1.2	1.7	1.3	1.5	0.9	1.8	0.7	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.9	79.6	86.2	92.2	98.5	105.4	105.2	102.9	99.3	92.6	82.2	71.4	(19)
		MCWB	49.2	50.3	54.9	56.2	60.6	64.4	68.4	68.1	65.9	60.5	53.8	48.5	(20)
	2%	DB	67.3	72.6	80.7	86.5	93.4	100.2	101.5	100.2	95.5	87.2	76.3	63.8	(21)
		MCWB	46.2	48.4	53.3	55.3	61.7	65.9	68.5	68.5	64.8	60.6	52.0	46.0	(22)
	5%	DB	60.5	66.7	74.3	82.0	89.6	96.6	99.2	97.0	92.0	82.1	70.4	59.1	(23)
		MCWB	43.3	46.1	51.3	55.8	61.3	66.7	68.7	68.6	65.7	59.0	50.4	43.7	(24)
	10%	DB	54.6	59.5	68.5	76.1	85.0	93.3	96.3	93.3	88.2	76.6	64.4	53.8	(25)
		MCWB	41.0	43.1	49.5	54.7	61.2	67.0	68.7	68.4	65.3	56.9	48.0	41.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	53.1	60.6	63.5	69.3	73.1	73.9	73.6	71.2	68.1	57.1	51.4	(27)
		MCDB	69.9	73.3	75.6	75.6	83.1	91.2	91.4	88.0	87.2	78.8	71.9	62.0	(28)
	2%	WB	47.3	49.4	56.8	60.4	67.4	71.4	72.2	72.4	69.4	64.7	54.2	47.6	(29)
		MCDB	63.8	69.9	69.4	75.0	81.6	89.3	90.4	88.2	85.9	75.6	69.2	63.1	(30)
	5%	WB	44.3	46.8	53.6	58.5	65.5	70.2	71.2	71.2	68.2	62.0	51.8	44.4	(31)
		MCDB	59.6	65.6	69.3	72.6	80.1	87.6	89.6	88.1	84.5	75.9	69.0	57.2	(32)
	10%	WB	41.4	43.9	50.7	56.0	63.5	69.0	70.3	70.2	66.9	59.3	49.1	41.7	(33)
		MCDB	53.9	59.3	67.7	72.0	78.4	86.1	89.3	87.0	82.5	73.0	63.7	53.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	27.5	28.3	29.8	28.9	28.2	27.9	27.2	25.9	27.4	28.2	28.9	25.5	(35)
		MCDBR	37.4	40.7	39.2	38.7	35.4	32.9	31.6	31.2	33.1	36.4	37.7	34.1	(36)
	5% WB	MCWBR	20.9	21.1	18.1	16.1	13.2	10.6	8.4	8.2	10.7	14.8	18.7	19.7	(37)
		MCWBR	35.6	38.8	34.1	29.9	29.1	27.3	27.2	26.2	26.5	28.5	32.9	32.2	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.257	0.265	0.293	0.322	0.341	0.373	0.378	0.364	0.337	0.300	0.271	0.256	(40)
		taud	2.577	2.567	2.492	2.428	2.436	2.385	2.412	2.420	2.479	2.551	2.561	2.582	(41)
	Ebn at Noon	Ebn at Noon	305	313	311	305	298	287	285	287	291	295	295	298	(42)
		Edh at Noon	23	27	32	36	36	38	37	36	32	27	23	22	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	908	1169	1569	1954	2232	2393	2334	2053	1747	1307	993	799	(44)
	RadStd	RadStd	82	113	132	108	133	117	101	105	83	142	89	56	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A
Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page